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Floating photovoltaic power plant: A review

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ABSTRACT

The noticeable rise in the electricity demand, fast depletion of fossil fuels, along with environmental concerns throughout the world has led to the requirement of commissioning Solar PV plants in large scale. Solar photovoltaic (PV) installation has the burden of intense land requirements which will always be a premium commodity. To conserve the valuable land & water, installing Solar PV system on water bodies like oceans, lakes, lagoons, reservoir, irrigation ponds, waste water treatment plants, wineries, fish farms, dams and canals can be an attractive option. Floating type solar photovoltaic panels have numerous advantages compared to overland installed solar panels, including fewer obstacles to block sunlight, convenient, energy efficiency, higher power generation efficiency owing to its lower temperature underneath the panels. Additionally, the aquatic environment profits by the solar installation because the shading of the plant prevents excessive water evaporation, limits algae growth and potentially improving water quality. This paper gives more insight about the Floating PV technology, its present status & various design options.

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1. Introduction

In recent years, renewable energy sources are growing rapidly all over the world. Solar energy is considered to be one of the most promising energy alternatives due to its ubiquity and sustainability. The solar energy is freely and enormously available throughout the world [1]. The most common application for the use of solar energy is all through the photovoltaic (PV) systems. Photovoltaic (PV) modules are one of the most effective, sustainable, and eco-friendly products in the field of renewable energy [2–7]. The installation of solar PV has the burden of intense land requirement which will always be a premium commodity. There is large water bodies available in various parts of the country which can reduce the saving cost of land and operating cost for power generation expenses [8]. So the solar PV systems can become a very logical alternative for harnessing solar energy by utilizing obtainable water bodies and help to increase the economic viability of solar projects. Energy from photovoltaic's though a renewable source, maintains a low efficiency of less than 15% in its long life use [9]. Floating solar generate more electricity than ground-mount and rooftop (solar) systems because of the cooling effect of water. It also reduces reservoir evaporation and algae growth by shading the water. The floating platforms are 100% recyclable, utilizing high-density polyethylene which can withstand ultraviolet rays and corrosion.

2. Types of solar PV installations

The classification of various solar PV Installations is shown in Fig. 1.

2.1. Ground mounted/conventional land based solar project

Ground mounted photovoltaic systems are generally large, utility-scale solar power plants. Their solar modules are held in place by racks or frames that are attached to ground based mounting supports. Ground based mounting supports include (Fig. 2):

- Pole mounts, which are single-minded directly into the ground or fixed in concrete.
- Foundation mounts, such as concrete slabs or poured footings.
- Ballasted footing mounts, such as steel bases or concrete that use weight to secure the solar module system in position and do not have need of ground penetration. This type of mounting system is well suited for sites where dig is not possible such as capped landfills and it simplifies decommissioning or relocation of solar module systems.

2.2. Roof top solar project

A rooftop photovoltaic power station, or rooftop PV system (Fig. 3), is a photovoltaic system that has its electricity generating solar panels mounted on the rooftop of a residential or commercial building or structure [10]. The various components of such a system include photovoltaic modules, mounting systems, cables, solar inverters and other electrical accessories. A rooftop photovoltaic power station (either on-grid or off-grid) can be used in conjunction with other power sources like diesel generators, wind turbine etc. This system is capable of providing a continuous source of power. Rooftop mounted systems are small compared to ground-mounted photovoltaic power stations with capacities in the megawatt range. Rooftop PV systems on residential buildings typically feature a capacity of about 5–20 kW, while those mounted on commercial buildings often reach 100 kW or more.

2.3. Canal top solar system

Conventionally Solar Plants are set up on ground requiring massive amount of land area. To avoid acquisition of large area of land, the new concept of setting Solar PV plant on Canal was conceived. By eliminating the use of land, not only deforestation is avoided but reforestation is encouraged through landscaping (Fig. 4).

2.4. Offshore solar PV system

Oceans cover more than 70% of the earth's surface; they receive a great amount of solar energy. The available solar resource could be exploited to counteract the current generation of electricity using solar PV technology. Due to the land scarcity onshore, the

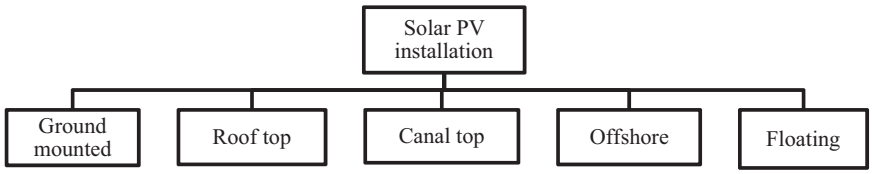

Fig. 1. Classification of solar installation.

Fig. 2. Ground mounted solar PV.

Fig. 3. Roof top solar power PV plant.

Fig. 4. Canal top solar systems.

Fig. 5. Offshore solar systems.

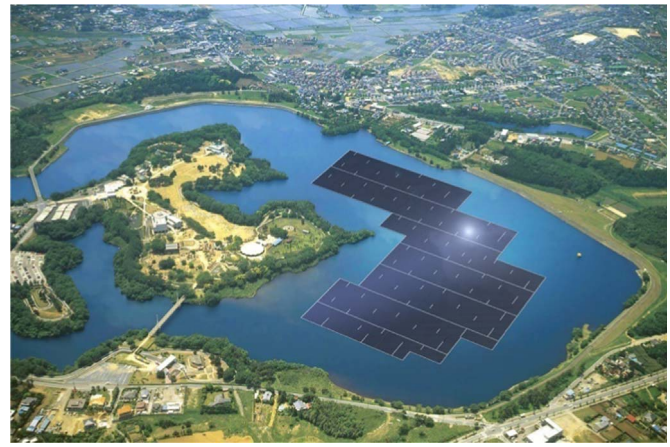

Fig. 6. Floating solar power plants.

offshore environment which takes full advantage of sun rays during the day is an ideal option for setting up PV plants (Fig. 5). Since one of the key components in PV panels is Cadmium Chloride, which is extremely toxic and expensive, it affects both the manufacturing process and the price of solar panels. The seawater contains Magnesium Chloride, which could replace the highly toxic and expensive Cadmium Chloride.

2.5. Reservoir/Lake based floating solar system

The Floating PV is a new concept, with no commercial deployments being undertaken and only few demonstrator projects being deployed worldwide [11]. There are many places around the world that do not have enough land for PV installations, mainly islands such as Japan, Singapore, Korea, Philippines and many others. There is already a demand for Floating PV in Japan, USA, Korea, Australia, Brazil, India and others. This demand is likely to increase and will spread all over the world. Floating solar systems can be installed in water bodies like oceans, lakes, lagoons, reservoir, irrigation ponds, waste water treatment plants, wineries, fish farms, dams and canals etc. A typical PV module converts 4–18% of the incident solar energy into electricity, depending upon the type of solar cells and climatic conditions. The rest of the incident solar radiation is converted into heat, which significantly increases the temperature of the PV [12,13]. The power output of solar cells varies according to change in temperature. Due to this efficiency of the PV module depend on the temperature so if we installed solar PV systems on the water surface benefit from a significant lower ambient temperature in virtue to the cooling effect of water [7,14–16]. If aluminium frames are used for supporting the floating solar PV module, it carries out the cooler temperature from water as well, bringing down the overall temperature of the modules (Fig. 6). On an average efficiency of floating type solar panels are 11% higher compare to ground installed solar panels [17]. A comparative advantages and disadvantages of the various Solar PV installations are listed in Table 1.

3. Concept of floating PV system

It is a new idea to install solar photovoltaic system over water bodies by using floating technology. The power generation results from the combination of PV plant technology and floating technology [8]. This technology replaces the installation of photovoltaic power plants over valuable land. The floating PV plant consists of a Pontoon or separate floats, mooring system, solar

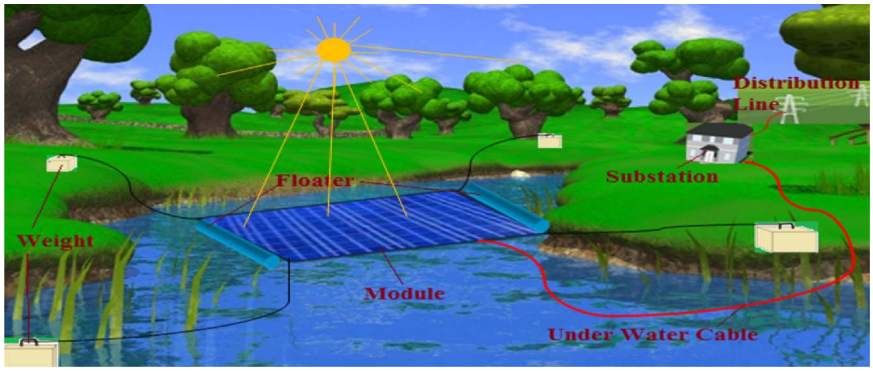

Fig. 7. Layout of floating solar power plant.

panels and cables (Fig. 7). According to a research, having this effective cover up from the pontoon and the PV panels on the reservoirs resulted in reduction of water evaporation from the reservoir. Research in Australia suggests that up to 40% of open reservoir's water could be lost during evaporation [18].

The most important parameter considered for the performance evaluation of the FPV is the PV effective conversion efficiency in operative conditions, which affects the electricity generation and thus the most valuable product of the component. The conversion efficiency of a PV module is given by the ratio between the generated electrical power and the incident solar radiation intensity, according to the following expression,

$$\eta el = \frac{Pmax}{SXA_{Pv}} x100\%$$

where η el is the electrical efficiency (%), Pmax is the power generated by PV module (W), S is the solar radiation intensity incident on the PV module (W/m^2) and A_{pv} is the front PV module surface exposed to the solar radiation intensity (m^2).

3.1. Components of floating PV system

- **Pontoon**: A pontoon is flotation device with buoyancy enough to float by itself as well as with a heavy load. The platform is design to hold suitable number of modules in series parallel combination according to the requirement and space availability [19,20]. The Fig. 8 shows the floats and the pontoon structure.
- **Floats**: Multiple plastic hollow floats with effective buoyancy to self weight ratio are combined over and over again, forming a giant pontoon. The floats are typically made of HDPE (high density poly-ethylene), known for its tensile strength, maintenance free, UV and corrosion resistance. Glass fibre reinforced plastic (GRP) can also be used for construction of floating platform. HDPE is commonly used for the fabrication of fuel tanks, milk bottles, water pipes, and can be recycled as well.

- **Mooring system**: A mooring system usually refers to any permanent structure to which a container may be secured. Examples include quays, wharfs, jetties, piers, anchor buoys, and mooring buoys. In the case of a floating solar system, the mooring system keeps the panels in the same position and prevents them from turning or floating away [21]. The installation of a mooring system can be a challenge and expensive in deep water. Mooring system for floating platform can be done with nylon wire rope slings which can be tied to bollards on bank and lashed at each corner. The Fig. 9 shows the mooring system which is used in floating power plant.
- **Solar PV module**: Till now standard crystalline solar PV modules have been used for the floating solar systems. However as more projects are installed on salty water surfaces, specifically fabricated modules will be required to resist the long term salt mist exposure. Nearly any metal will corrode over time and therefore alternatives to standard aluminium frames and mounts, such as polymer made frame is needed.
- **Cables and connectors**: Electricity is drawn from the solar array and transported to the land. Therefore, the power can be fed to the grid or stored in batteries. The projects commissioned so far, did not have cables pulled under water, but kept wiring above water. Even though no electrical components are under water, properly rated cables and waterproof IP67 junction boxes are important with floating solar projects. Other electrical components such as inverters and batteries remain 'nice and dry' on the land. High temperature resistance, water proof and robust cables are to be used to provide a long service.

3.2. Key design factors

3.2.1. Layout of pond/lake/reservoir

Floating cover systems require site specific planning and design to be successful. In addition, both the reservoir's walls and the different design layouts for the internal 3D geometry of the reservoir are highly incompatible. As a result, the geometry of the floating module has to be adaptable enough to suitably get used to different internal geometries of the water reservoir.

3.2.2. Floating structure/geometry

The floating module's geometry was designed taking into account two main issues. First, the dimensions of the module must be modified to commercial photovoltaic panels. Second, the modules must cover the maximum possible water surface to avert water evaporation [22]. The solar issues under analysis were: photovoltaic panel dimensions and tilt angle, number of units to be installed, distance between panel rows to prevent shade effects and access ways to ease operational maintenance.

3.2.3. Orientation of floating PV panel

India is lying in northern hemisphere with latitude of 22° north and longitude of 77° east. The main longitudinal axes of the reservoir are to be aligned with the cardinal directions and the solar panels should face south [22]. In some cases the panels could be designed to not be oriented and so not having any particular orientation.

4. Economical analysis of floating solar power plant

The economics of 1 MW Floating solar Power plant has been

Table 1
Advantages and disadvantages of various floating solar PV installations.

Ground mounted	Roof top	Canal Top	Off Shore	Floating solar
Advantages: • More scope and lower cost to install a sun tracking system. • Facility to operate a manual seasonal tilt adjust system. • Larger systems can be installed because additional space on the ground then on the roof in rural settings. • Panels are easier to clean and maintain. • It also facilitates to avoid 'voiding roof warranties' as no connection between the system and the roof is involved.	• Aesthetics: the panels fit impeccably with the pre-existing rooftop for a more efficient appearance. • Space optimization: with rooftop solar, there's no need to clear away extra land. • Fortification: solar panels shield roof and protect from weather and wear-tear. This, in turn, increases the lifetime value of property. • Speed: rooftop solar is usually easier and faster to install than ground-mounted systems.	• Save precious and costly Land. • Save canal water from evaporation. Generate power with higher efficiency compared to land based solar power plants due to cooling effect on solar panels by evaporating canal water. • Extended Service Life and Energy Gain (Reduced Degradation of Semiconductor).	• Due to direct contact of PV panels with water, the negative coefficient efficiency (%/K) of the PV junction can be used to generate a higher electrical acquiesce for the same area occupied onshore; • As the PV panel temperature falls, the efficiency of panel increases. • In hot conditions it solves dual purpose of power generation and prevents water from evaporating from beneath. • Takes full advantage of sun rays during the day.	• **Increased Efficiency:** Reflection of light from the water and the natural evaporative cooling as an outcome of the water body can maintain the PV panel temperatures lower than land based ones and hence boost their efficiency. • **Reduced Evaporation of water**: Floating PV system provides shading to the water surface and reduces evaporation. • **Improved Water Quality**: It can also lead to better water quality due to reduced photosynthesis and algae growth. • **Less dust effect**: Typically areas with high solar energy potential lean to be dusty and arid, so in comparison to their ground mounted counterparts, floating PV systems perform in a low dust environment. • **Land saving:** It saves precious land for agricultural, mining, tourism and other land-incentive activities and turn unexploited and non-revenue generating water surface into commercial solar power plants.
Disadvantage: • Urban settings often do not have enough land space. • Solid foundations and concrete footing will need to be built to provide stable structure protecting from storms and high winds. • Construction time is more because of Civil and engineering work than other system.	• Roofs may have too many obstacles like chimneys, trees, vents, satellite dishes, etc.) Leading to shading losses. • Roof may not properly fit to the required system capacity. • Lack of Southward facing roof which may affect yield • Labour intensive.	• Lack of availability of Canals for such projects. • Socio economic and political issues of using rivers and canals. • Complicated and lengthy structures to accommodate modules. • Maintenance of such systems is an issue due to sub-optimal location. • Local shading by trees surrounding canal which cannot be uprooted for ensuring soil's stability and preventing erosion. • Covering such canals with PV panels essentially destroys the wetlands for birds and greenery areas. • Panels, structure etc. may lead to contamination issues of fresh water. • Better (structural and design strategies is required which will increase the cost. • Power evacuation of small capacity over a long path would be difficult and costly due to increased Cable costs. • System spread over such a large length cannot be protected by boundary walls or fencing, thus security concerns are very major in such cases.	• Since one of the key components in PV panels is cadmium chloride which is expensive and enormously toxic, it affects both the manufacturing process and the price of solar panels. Researchers found that seawater contains magnesium chloride, which could replace cadmium chloride. • The panels are designed to be waterproof. • Panels are to be made lighter so that they can float which require high cost material or otherwise we have to use some structure which makes whole installation costly. Connecting of solar panels and their maintenance in water as well as connecting to the grid could be a major issue for this type of plant.	• The system is prone to more threats like High tides, storms, sea waves, cyclones and tsunami. • Increased corrosion of the metallic structure and components which can reduce the life of the system. • Causes reduction in penetration of the solar lights into the water bodies which may affect the growth of aquatic animals and seaweed etc. • Reduced Humidity and Temperatures on the panel cause negative thermal drift which may decreases the overall efficiency. • Regular Cleaning of clay accumulated on the sides of the river/lake is needed. • Fishing and other transportation activity may be affected depending the selected site.

Table 2
Review of various floating solar PV installations worldwide.

S. no.	Company name	Capacity	Location/place of installation	Reference
1	Kyocera TCL solar	2.3 MW	Hyogo prefecture, western Japan	
2	Kyocera TCL solar	1.7 MW	Hyogo prefecture, western Japan	
3	Kyocera TCL solar	1.2 MW	Higashira pond, Japan	
4	Infratech Industries	4 MW	Jamestown, South Australia	
5	Kyocera TCL solar	13.4 MW	Yamakura dam, Japan	
6	SPG Solar	175 kW	Napa valley's Far Niente Wineries, California, (US)	[25]
7	National institute of advanced industrial science & technology	20 kW	Aichi, Japan	[26]
8	SPG Solar	30 kW	Gundlach bunshu wineries, California, (US)	[27]
9	Bryo	500 kW	Bubano, Italy	
10	Celemin energy & polytechnic university of Valencia	24 kW	Agost, Spain	
11	D.A.I.E.T	20 kW	Solarolo, Italy	
12	Tera Moretti Holding	200 kW	Petra Winery, Italy	[28]
13	Celemin energy & polytechnic university of Valencia	300 kW	Agost, Spain	[28]
14	SCINTEC	30 kW	Lake colignola, Italy	[28]
15	D.A.I.E.T	20 kW	Avetrana, Italy	[28]
16	Ceil et terre	14 kW	Piolenc, France	
17	SPG Solar	350 kW	Petaluma, California (US)	
18	Osesol	4 kW	Vendee, France	
19	ENERACTIVE	112 kW	New Jersey, (US)	
20	Techwin	20 kW	Cheongju, South Korea	
21	K-Water	500 kW	Hapcheon Dam	[29]
22	Osesol	100 kW	Pommeraie -sur - sevre, France	
23	Phoenix Solar	5 kW	Bishan Park, Singapore	
24	MIRARCO	0.5 kW	Sadbury, Canada	
25	Ceil et terre	1.157 kW	Okegawa, Japan	
26	Ceil et terre	200 kW	Sheeplands farm, Barkshire, UK	
27	Kyocera	7.5 kW	Umenokifurukori reservoir, Japan	
28	Vikram solar Pvt. Ltd.	10 kW	New town, West Bengal, India	
29	NHPC Ltd.	50 MW	West Kallada, Kerala India	
30	MANIT	1 kW	Bhopal, M.P., India	

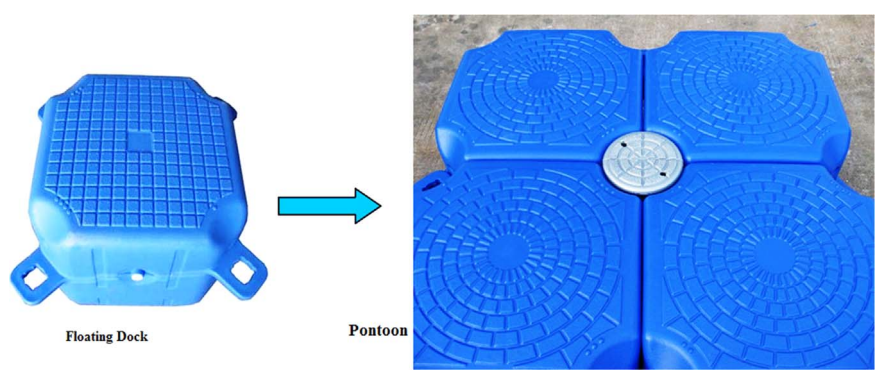

Fig. 8. Pontoon Structure.

Fig. 9. Floating active cooling and concentrating design, SIT – Italy.

worked out and the payback period is calculated. The payback period of the plant is only 5 years based on calculation and plant life will be minimum 25–30 years. [23]

$$\text{Payback period} = \frac{\text{Total cost of PV system with all auxilliary equipment}}{\text{Total annual cost saving after installation of PV system}}$$

4.1. Calculation of payback period

Total No of Module Nos. 4000.
Capacity of each Module Watt 250.
Total Capacity of the Plant MW 1.
Installation Cost for 1 MW Unit crore 8.
Total Installation Cost crore 8.
Selling Cost per Unit Rs/ kWh 9.
Total Generation Hrs Hrs 1920.
Total Generation (MU) MU/per Day 0.008.
Total Generation (MU) MU/per Year 1.92.

Earning Per Year Crore 1.728.
Savings after 5 years Crore 0.64.
Savings after 10 years Crore 9.28.
Pay Back Period is 5 year (approx.).

5. Commercial designs of Floating Solar System

5.1. Photovoltaic floating rotating active cooling and concentrating solar system

The floating solar plant is constructed to float on a raft casing that is free to track the sun and takes benefit of the cooling properties of the water body. This systems installed on the water surface benefit from a significant lower ambient temperature due to the evaporative cooling effect of water. The aluminium frames certainly conduct the cooler temperature from the water as well, bringing down the overall temperature of the modules. The system can operate under all weather conditions and also withstand seismic loads. In this design, it also includes floating solar unit combined with cooling, tracking and concentrators to gain maximum solar energy. System allows exploiting basins, natural and artificial lakes to install PV plants. This system consists of a series connected floating rafts with PV panels supported by tubular frame buoyant base. The power of a single PV module ranges from 1 W to 300 Wp, depending on the type of system configuration and panel used. The modular structure allows different plant sizes and configurations: fixed installation (in order to maximize the coverage of the available area) or tracking installation (in order to maximize the energy collection) [24].

5.2. Colignola Pilot Plant

The Floating Tracker Cooling Concentrator (FTCC) uses reflectors to increase the efficiency of the solar energy capture during different times of the day. The FTCC also uses water to help stay cool, reduce costs to about 20% less than conventional, ground based systems [24]. The reflection of the water surface and the cooling effect of the surrounding water can increase the electricity yield from the solar panels in FTCC systems (Fig. 10).

5.3. Submerged photovoltaic solar panel

In this configuration the panels are immersed in water and this allows us to realize a gain in efficiency of approx. 20% in summers compared to a normal panel exposed to the air (Fig. 11).

Fig. 10. Floating Tracker Cooling Concentrator FTCC Design, SIT – Italy.

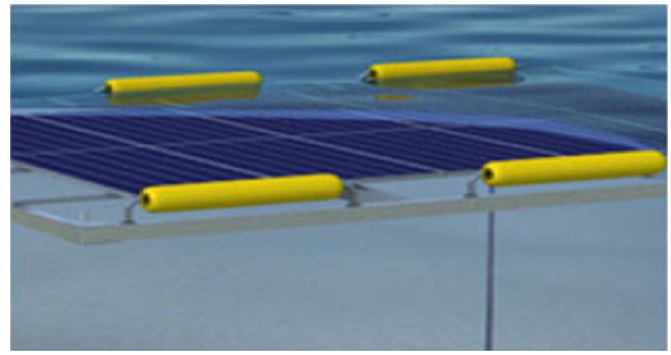

Fig. 11. Submerged photovoltaic design, Infratech Industries Inc., Australia.

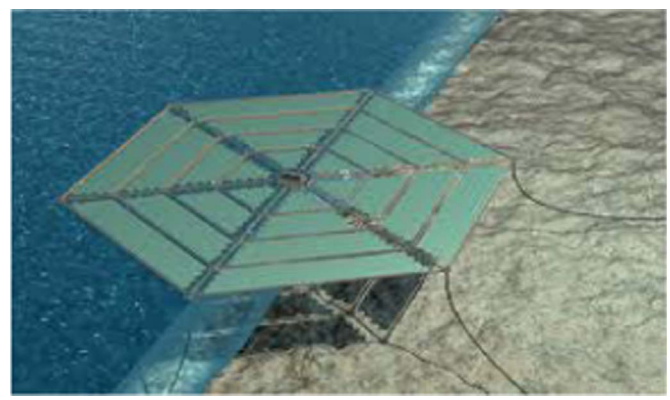

Fig. 12. Flexible floating hexagonal PV Design, SUNdy concept.

5.4. SUNdy concept, the hexagonal design

The other flexible floating PV concept is called SUNdy (2012), realised by Det Norske Veritas (Fig. 12). This design consists of a series of thin film PV panels connected together and then onto the electrical bus lines running through the hexagonal vertices. The panels themselves are envisaged to be laminated and adhered to a flexible foam surface, which gives the panel's buoyancy and structure. At the edge of the float is embedded a marine grade connector, which allows the panels to be connected both mechanically and electrically. A transformer is placed at the centre of the hexagonal structure from which the electricity is delivered to shore. Plans are for walkways and water cannons, for cleaning of the panels, to be located between the centre and the vertices to allow contact to the equipment. The structural design is stirred from a spider web and is designed to be submissive with the waves while being structurally strong and capable of maintaining its shape. The whole structure is designed to be kept back in place.

5.5. HYDRELIO© Floating solar components

The latest such 'mega plants' at Nishihira and Higashihira Ponds in Kato City are the works of Kyocera Corporation and Century Tokyo Leasing Corporation, that took only seven months to complete the installation (Fig. 13). The plant's 11,250 modules are expected to generate 3300 MWh every year. According to Kyocera Company, besides being typhoon-proof (due to their sturdy, high-density polyethylene and array design) floating solar PV plants are superior to their land-based equivalents because of the cooling effect of water, which allows them to function more efficiently.

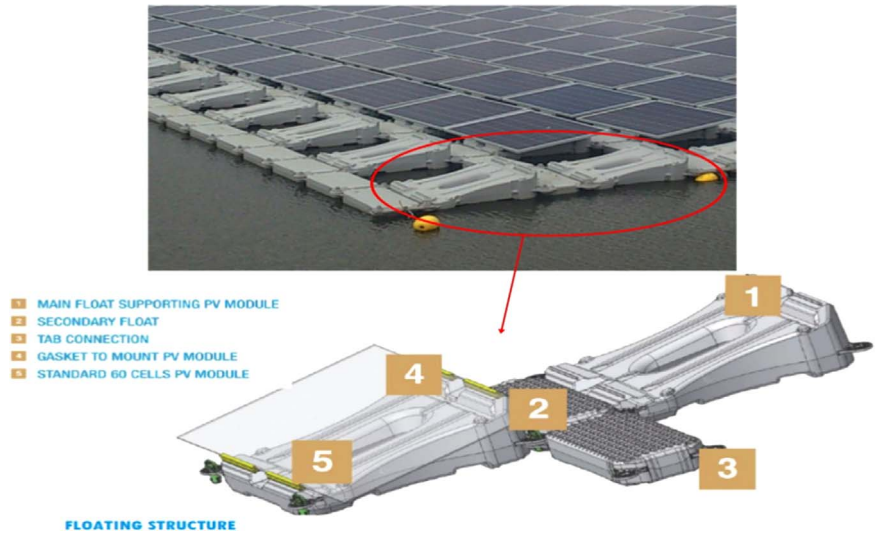

Fig. 13. Floating solar technology Design, Ciel & Terre's, France.

6. Review of various floating solar PV installations worldwide

There are very less number of manufacturers involved in development of floating PV system worldwide (Table 2). In India also, manufacturers of PV are planning to enter the market of floating PV. India is currently making plans to build the world's largest floating solar power plant. The plant is expected to produce 50 MW. This major project is expected to cost between 64 and 72 million dollars. It will be set up in bodies of water in the Southern state of Kerala by the National Hydro Power Corporation with technical assistance from MANIT Bhopal.

7. Floating solar PV systems Vs land based systems

7.1. Benefits

- Floating type solar photovoltaic panels have numerous advantages compare to conventional solar panels, including convenient, and energy efficiency. Floating type solar photovoltaic panels have higher power generation efficiency owing to its lower temperature underneath the panels compare to overland installed solar panels [29].
- Shading effect, reduction in algae growth, natural reflectivity of the water surface, reduced sunlight penetration; Lower water temperatures have positive impact on the performance of Floating Solar PV [30].
- Reduce water evaporation, conserve water by lowering the temperature of water and reducing the size of water area exposed to air, floating solar panels can reduce water evaporation by up to 33% on natural lakes and ponds, and by about 50% on man-made facilities [31].

– Save precious land for agricultural, mining, tourism and other land-incentive activities and turn unexploited and non-revenue generating water surface into commercial solar power plants. Technology can lead to considerable savings on land prices and bring down power generation expenses.

- Floating solar system offers a holistic approach for inland freshwater bodies, Remote Island, Hydroelectric dams,industrial ponds, quarry and mine lakes, irrigation reservoirs and water treatment sites to become solar-friendly real estate
- Floating solar is cost competitive with roof and ground-based single-axis tracking solar systems and uses the same commercially available solar panels.
- Floating solar installations in most countries qualify for federal subsidies, grant and incentive programs similar to land-based solar.
- Water for cleaning the panels (& hence increasing efficiency) is readily available. The benefit that water gets from the installation of solar panel above the water surface is also contributing to the widespread acceptance of the floating solar panels [32].

– The installation is relatively easy to implement, since the floatation structure can be assembled without heavy equipment. Considerably reduced installation time and associated costs due to very limited site preparation needs [33].

7.2. Challenges/Issues

The biggest challenge in the Installation of the floating solar project is the system design which has to be suitably designed to stay afloat and be able to withstand the force. The following challenges are to be addressed during the installation of floating solar power plant.

- The solar modules are surrounded with water due to the system performance may be affected due to high moisture contents [34].
- The strength of the floating structure may be affected because of corrosion and adverse environmental condition [35].
- Safety issue in transporting the power from the water surface to the land area.
- The Floating system should be able to deal with environmental factors such as water quality, Varying water depth, temperature variations, water current, temperature, evaporation, oxygen, fish, algae growth and other live organisms.
- Floating solar systems may encounter rapid or erratic movement owing to floods, cyclone, waves and heavy winds. The floating PV system needs to be able to withstand these forces of nature.
- The high initial installation cost in addition with high maintenance cost are two of the major restrain for the expansion of the floating solar panel market.
- The power generation cost from solar panels is about 10 times costlier than the other fossil fuel based technique during its initial years [35].

- The installation of floating solar panels cannot be done in sea as the sea tide affects the position of floating solar panels continuously [36].
- The high wind speed in sea also affects the power generating efficacy of the floating solar panel system.
- Floating solar plants need directional control mooring systems to maintain effectively same azimuth (direction) and position on the water. Because directional change of solar modules reduces power output.
- Stress and vibration issues are more common in floating solar plant because of wind, waves and external forces. Vibration may lead to micro-cracks formation in modules which in turn reduces the electricity production and durability issues.
- The engagement of the public and relevant organization in the early stage of planning, in order to ensure public acceptance [18].

7.3. Cost

The support structures used for floating plants can account for up to 25% of total project costs but this amount is often less than the cost of buying and preparing an equivalent area of land nearby. Relatively high costs of land, civil works and seismic-proof foundations are needed in land based Installations. Operation and maintenance costs are also often reduced compared to land-based systems because the water needed for cleaning is available at source and components were less likely to overheat. Saltwater corrosion is not normally a problem since most floating PV is sited on freshwater bodies such as lakes and reservoirs. In addition, most balance-of-system equipment is usually sited on shore and is easy to access. Floating PV is potentially less prone to shading and there is no maintenance associated with clearing away ground-based vegetation [37].

7.4. Encapsulations for different climatic Zones for water floating systems

Solar PV modules require different encapsulations according to the local climatic conditions [38].

7.4.1. Designing for ice

Coping with ice is a particular wrinkle in floating PV systems for many regions. Specially designed floating installation to withstand heavy ice/freeze/thaw environment is required which will rise and fall with the reservoir water level. Solar “water bees” can be used to churn the water at the reservoir to improve water quality.

7.4.2. Concentrating PV cools in pools

Apart from reflective enhancements to floating PV systems, concentrators are also under development. “The great advantage of concentrated PV (CPV) technology in general has numerous opportunities for cost reduction. Synchronization of modules to the daily and seasonal movement of the sun and the cooling demonstrates improved power output over other commercially available. CPV systems can operate between 25–55 °C and higher. Water can also be used a structural support and the system may be made neutrally buoyant without use stainless steel or concrete or posts in the ground. Traditional concentrated photovoltaic technology consisting of a plastic lens and plastic concentrators can be used to track the sun.

7.5. Certification test for floating installations

Floating PV systems experience different dynamic stresses compared to those encountered by standard ground-mounted installations. Therefore rigorous component and panel evaluations have to be performed in real and simulated floating conditions. This includes component salt spray, panel vibration, corrosion, oxidation, immersion and UV exposure tests. Military grade solar modules are needed for ocean installation. These modules undergo a Salt Mist Corrosion test according to the IEC 61,701 standards for special certification.

8. Environmental impacts of floating solar PV plant

- May create an impact on ecologically protected and susceptible areas.
- Potential reduction in algae growth due to reduced sunlight diffusion and reduced photosynthesis.
- The silicon modules and High-Density Polyethylene (HDPE) thermoplastic floats; may affect the quality of water
- Possibilities of electrical accidents owing to underwater cables and have impact on existing ecosystems.
- Fishing and other transport activities in water bodies may get affected.
- Biodiversity of aquatic system may likely to get affected.

9. Floating solar development forecast to 2020

Asia pacific is the largest and fastest growing market of floating solar panel followed by Europe, Japan, China and India. A new market opportunity lies in the expansion of floating panel type solar power system in densely populated countries such as China, India, Japan, USA, Korea, Australia, Brazil and others where there is shortage of land that can be used for the installation of overland solar panels. As the cost of water surface is much lower than the cost of land the demand for Floating PV is expected to increase and will spread all over the world. In addition arability of the water resources, demand and supply gap of electricity in India and china are also high in comparison to the developed countries which in turn expected to boost the market of floating solar technology. India has taken a challenge of installing of 100 GW capacities of solar power installation and generation by the year 2022. In India, such technology can contribute to the share of the RE based generation targets and save the limited water sources availability.

10. Concluding remarks

This paper highlights the concept of floating PV system installed on still water bodies such as ponds, lakes, dams and reservoirs. It also compares the installed capacity of floating PV plants across the world. The following conclusions are drawn from the study.

1. A floating solar technology would prove to be an innovative step as it could solve the perennial problem of land.
2. As these solar panels would be floating on water, they are expected to stay cool and hence generate more power than those set up on land.
3. In India large water bodies are available in eastern, Sothern and South-eastern part of the country in states such as West Bengal, Assam, Orissa and Andhra Pradesh, Tamil Nadu and Kerala. This technology can be adopted in these states leading to considerable savings on land prices and bring down power generation expenses, thus reducing the gap between thermal and solar power.
4. Continued research on designing anchoring system for floating

PV system is needed to completely fix the buoyancy system.
5. The effect of salt water on the PV structure and the module performance has to be researched.
6. Development of solar tracking system that can change the tilt and azimuth angle of floating PV system is required.
7. Most of the projects in existence incorporate rigid crystalline PV modules which are incapable of withstanding harsh water environment therefore research on flexible thin film technology for such harsh condition have to be explored.
8. Developments of large megawatt scale floating solar farms in near future may pay way for the offshore solar technology development.
9. Maximum speed of wind, water current, temperature limit, snow load, cyclone and typhone has to be considered while designing the solar panel.
10. The efficiency of floating solar plant is 11% higher and reduces the water evaporation by 70%, however the investment of such power plant is 1.2% times higher than the conventional solar power plant.
11. Remote sensing and GIS based techniques can be used to determine the potential of floating solar PV projects.
12. Appropriate safety measures to transport the power from the water bodies to the land have to be carried out.

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